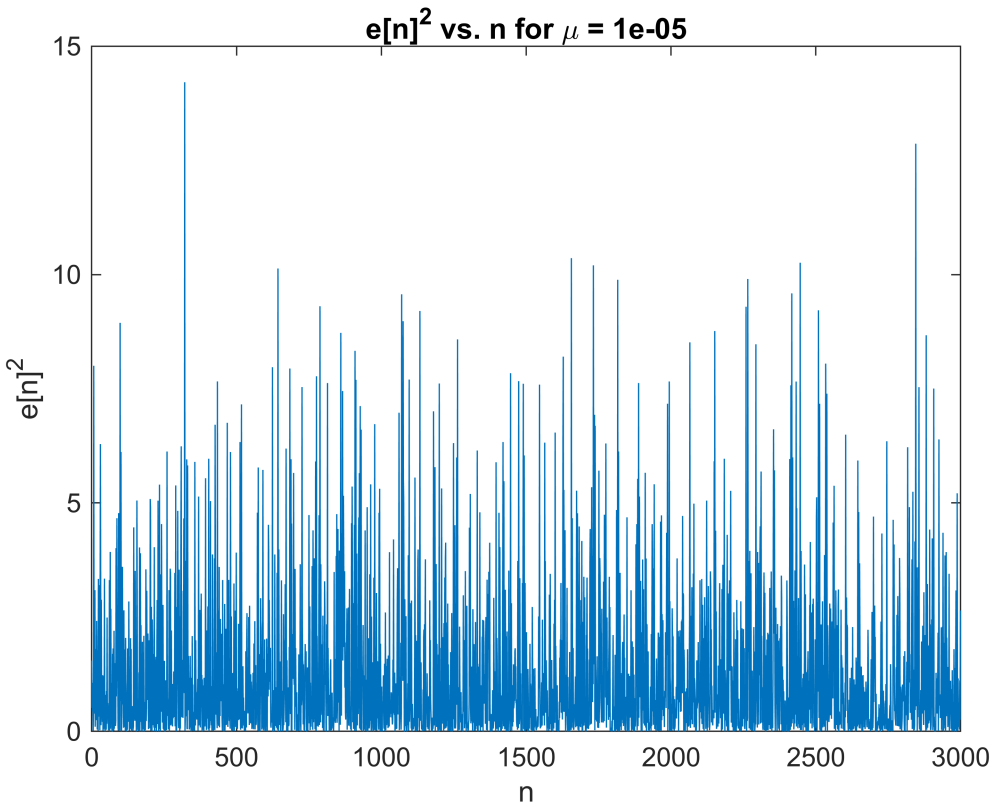
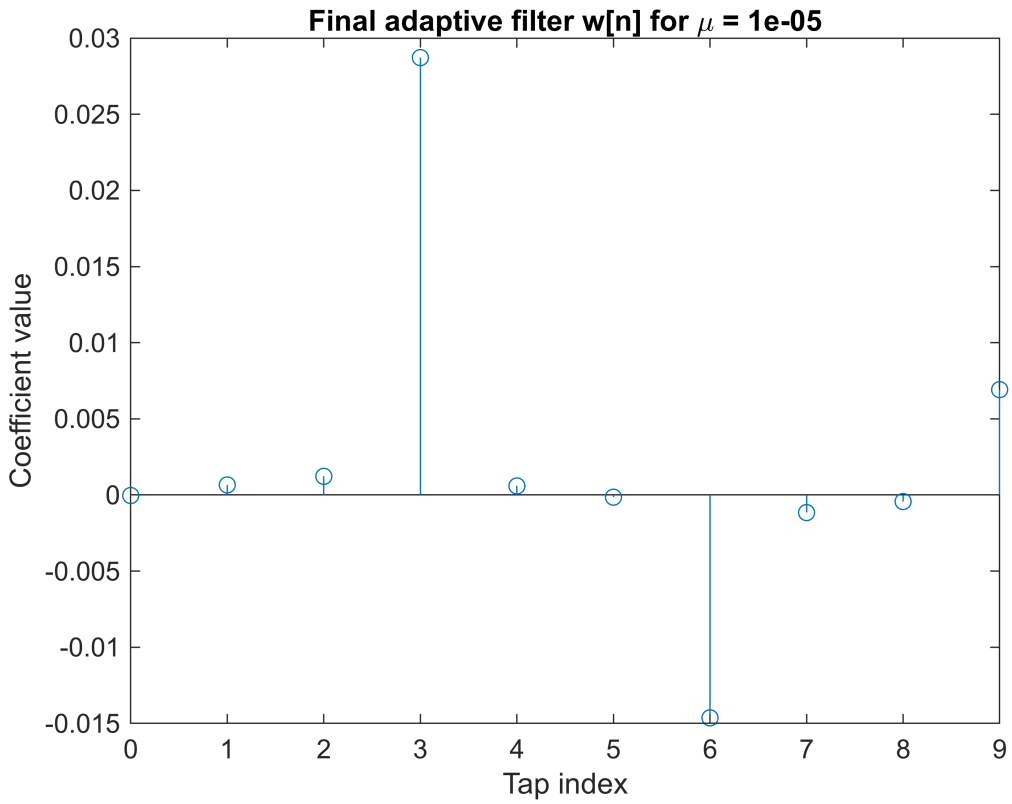
6) (15 points) Implement an adaptive filter in the System Identification architecture using MATLAB to match the output of an unknown system h = [1, 0, 0, -0.5, 0, 0, 0.25]. Use a filter length of 10 for the adaptive filter w and use the MATLAB function randn(1,N) to create the input sequence for both the unknown system and the adaptive system. N should be set equal to the number of output samples needed plus (M-1) where M is the filter length. The desired signal will be the convolution of the unknown filter h and this input sequence.

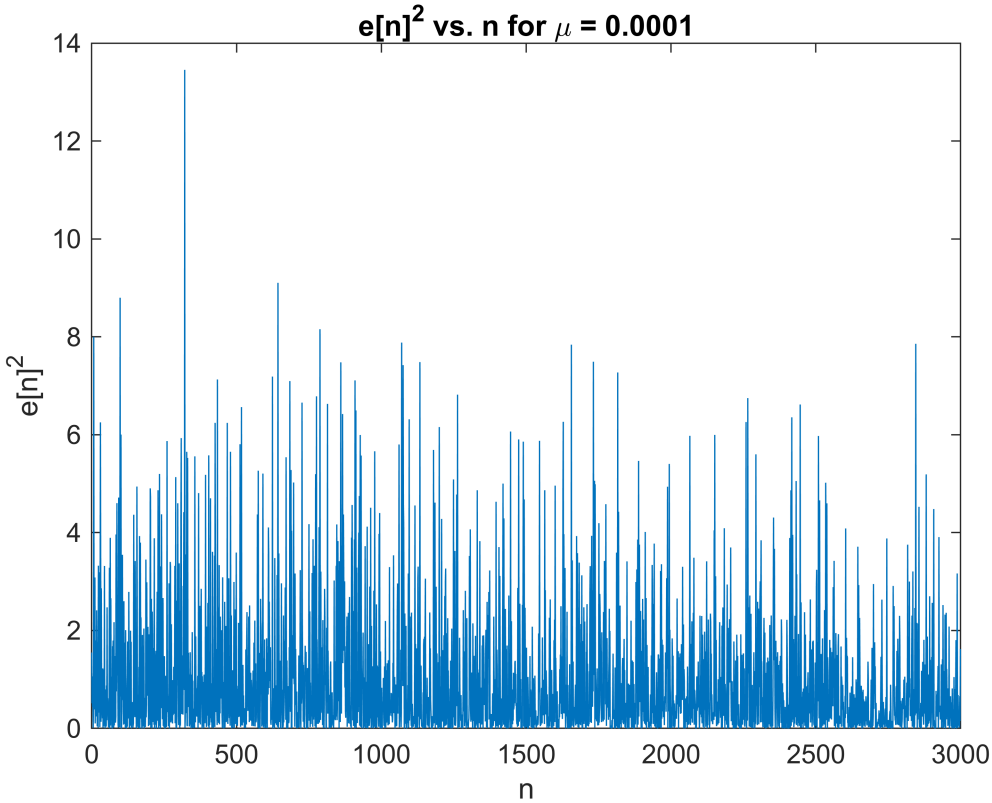
a) For three separate adaptation constants, $\mu = 10^{-5}, \mu = 10^{-4}, \mu = 10^{-3}$, adapt the filter for 3000 samples of the desired signal. For each adaptation constant, plot values of $e[n]^2$ for $0 \leq n \leq 2999$, and also make a stem plot of the adaptive filter values after 3000 updates.
b) Compare the final value of w[0] of the three adapted filters to h[0] and to each other. How is the final value of w[0] related to the adaptation constant? What would you expect the value of w[0] to be for $\mu = 10^{-6}$?
c) For each of the three values of the adaptation constant, compare the general distribution of values of $e[n]^2$ for n near 3000 to the initial range of values of $e[n]^2$ for n near 0. How is this related to your answer to part (b)?

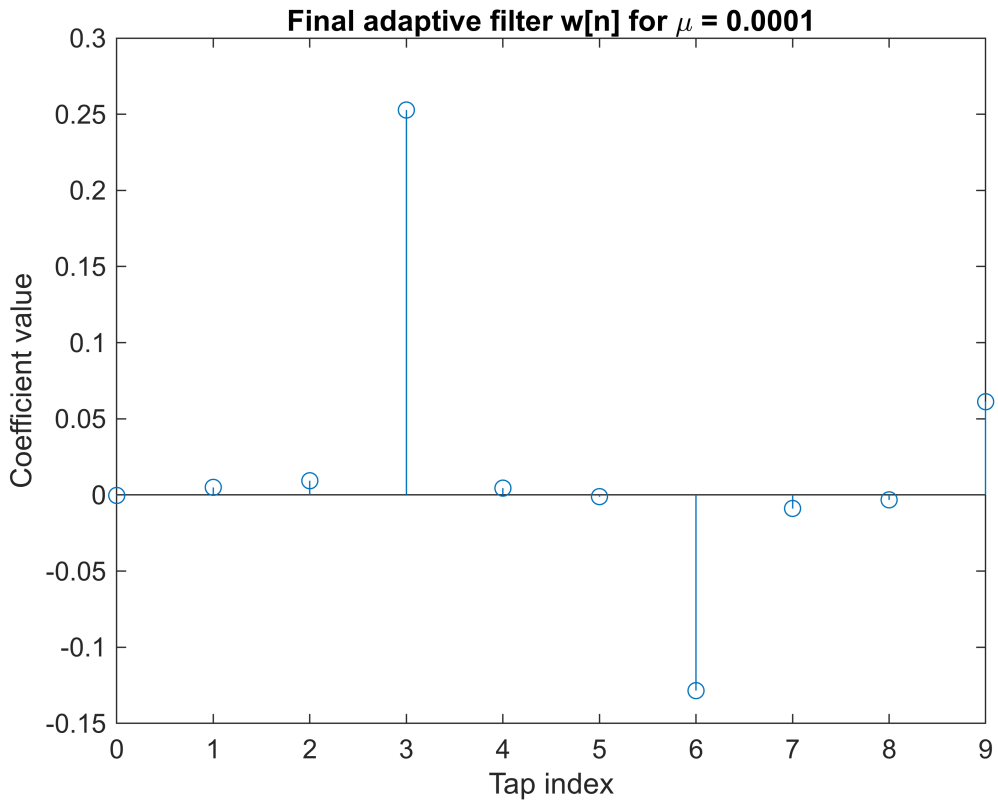
```
h = [1, 0, 0, -0.5, 0, 0, .25];
M = 10;
mu_vals = [1e-5, 1e-4, 1e-3];
N = 3000 + M - 1;

x = randn(1, N);

d = conv(x, h, 'valid');  % length = 3000

for k = 1:length(mu_vals)
    mu = mu_vals(k);

    w = zeros(M, 1); % adaptive filter
    e = zeros(1, 3000); % error
    y = zeros(1, 3000); % output


    for n = 1:3000
        x_n = x(n+M-1:-1:n).';
        y(n) = w' * x_n;
        e(n) = d(n) - y(n);
        w = w + mu * x_n * e(n);
    end

    % plot error squared
    figure;
    plot(0:2999, e.^2);
```

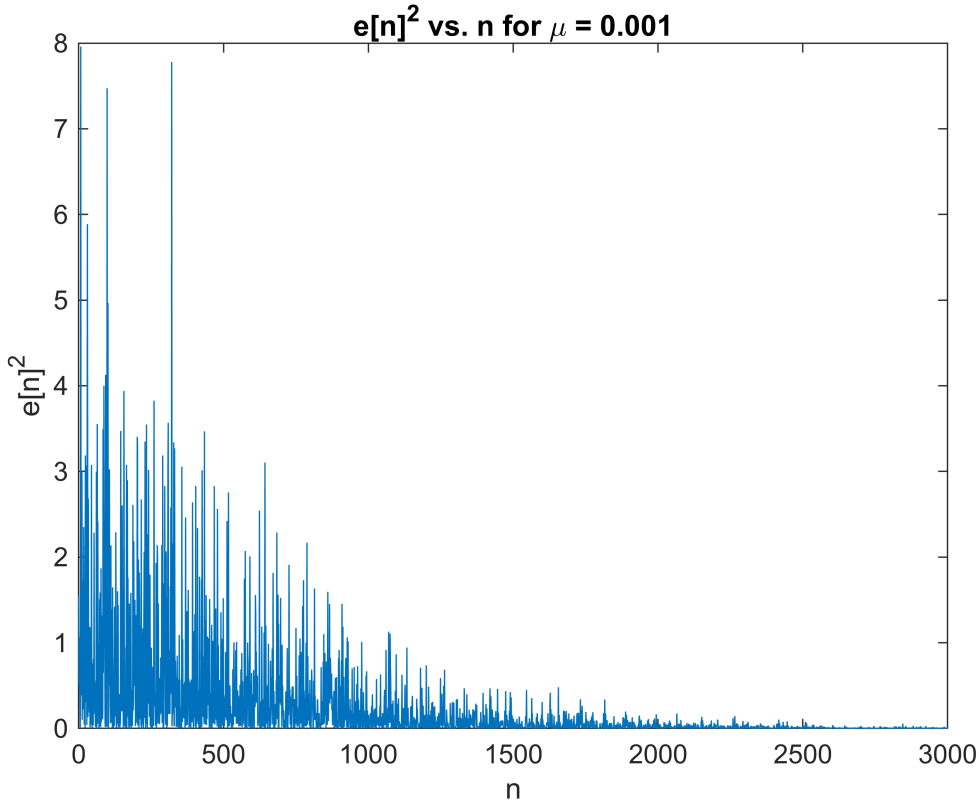
```
    title(['e[n]^2 vs. n for \mu = ', num2str(mu)]);
    xlabel('n');
    ylabel('e[n]^2');

    % stem plot of final filter coefficients
    figure;
    stem(0:M-1, w);
    title(['Final adaptive filter w[n] for \mu = ', num2str(mu)]);
    xlabel('Tap index');
    ylabel('Coefficient value');
end
```

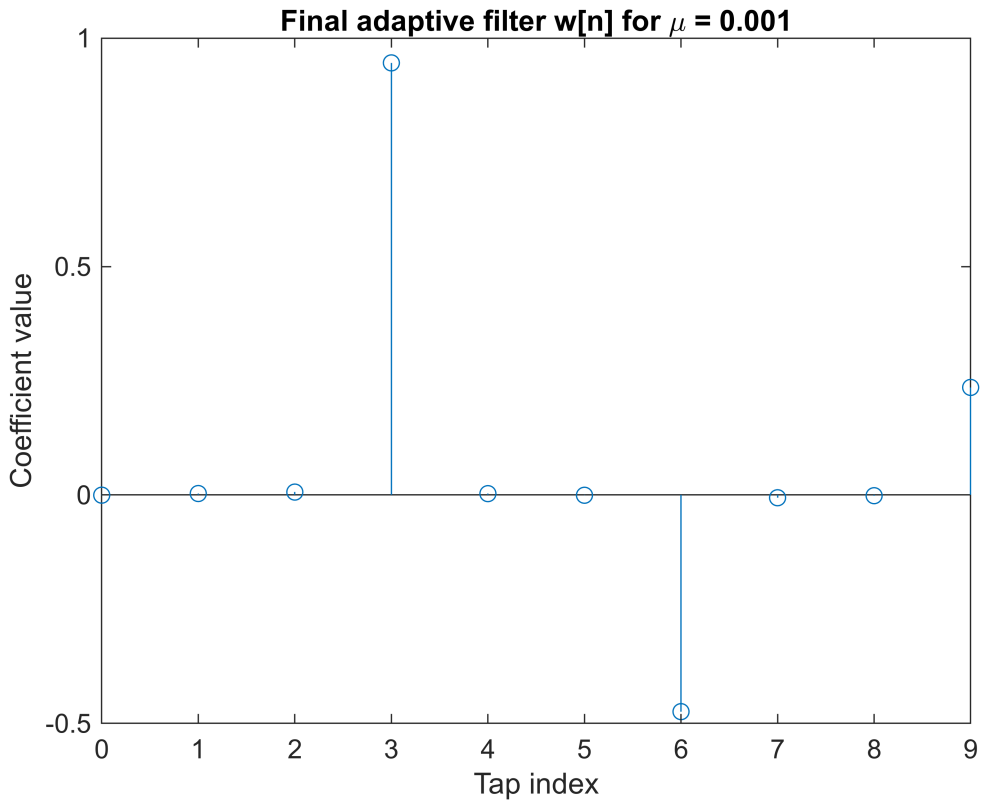



Final adaptive filter w[n] for μ = 1e-05
Coefficient value
Tap index
0.03
0.025
0.02
0.015
0.01
0.005
0
-0.005
-0.01
-0.015
0
1
2
3
4
5
6
7
8
9


e[n]^2 vs. n for μ = 0.0001
e[n]^2
n
14
12
10
8
6
4
2
0
0
500
1000
1500
2000
2500
3000


Final adaptive filter w[n] for μ = 0.0001
Coefficient value
Tap index
0.3
0.25
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
0
1
2
3
4
5
6
7
8
9


e[n]^2 vs. n for μ = 0.001
e[n]^2
n
8
7
6
5
4
3
2
1
0
0
500
1000
1500
2000
2500
3000



mu=.001 gives us the closest w[0] value as it's almost 1, while the others are ~.25 and ~.03. A higher mu leads to better adapting. I would expect w[0] to be even lower for mu = 1e-6.

We also see as mu increases, the error function drops off way more. It's really only visible for mu=.001.

7) (10 points) The stable range of the value of the adaptation constant is related to the power level of the input.

a) If the input is a*randn(1,N) instead of randn(1,N), how will this affect the outer product $x_n \cdot (x_n)^T$?

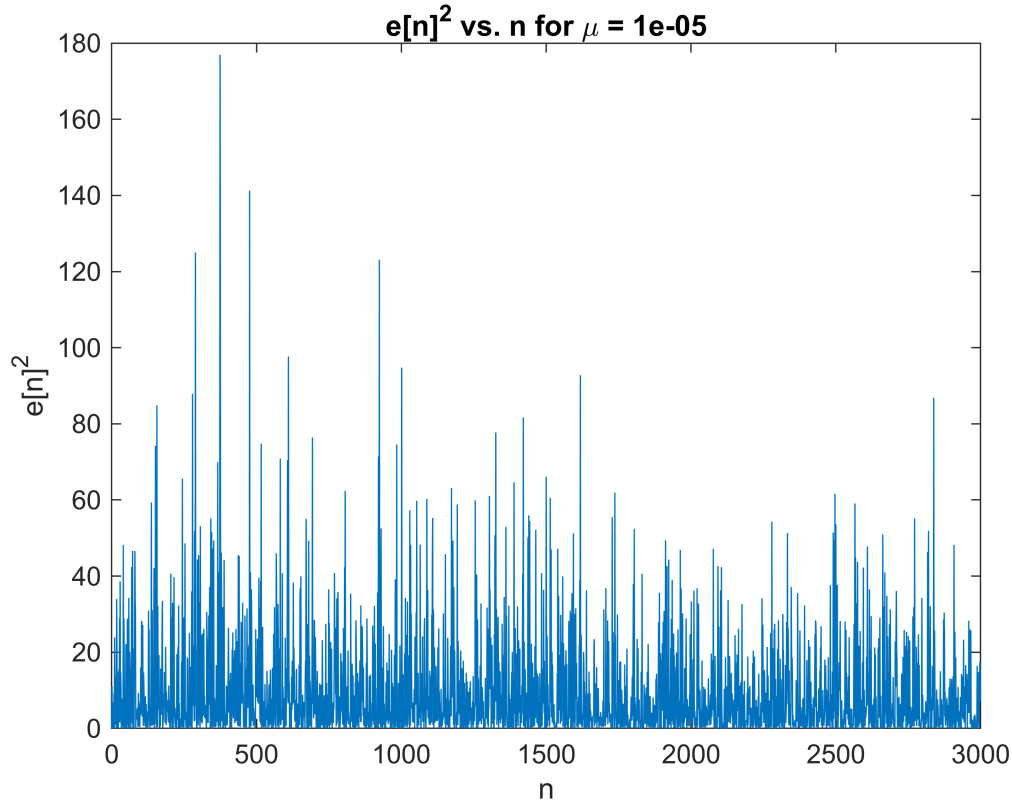
The outer product is now axn * (axn)^T = a^2 * (xn * (xn)^T)

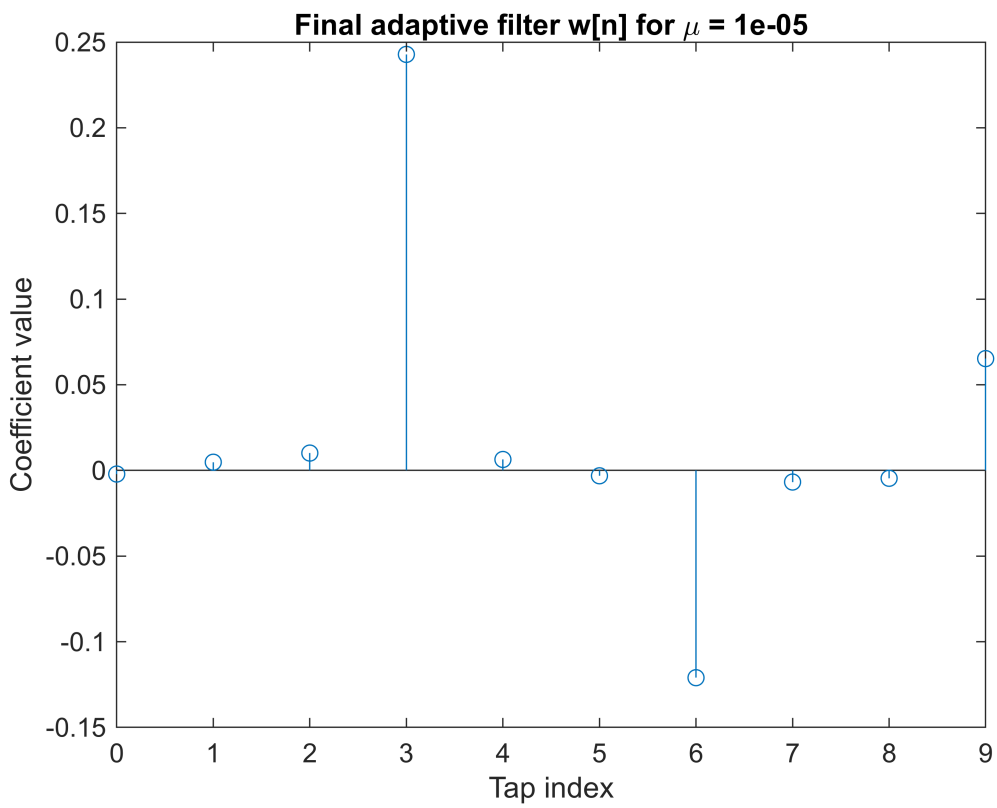
It gets scaled by a^2.

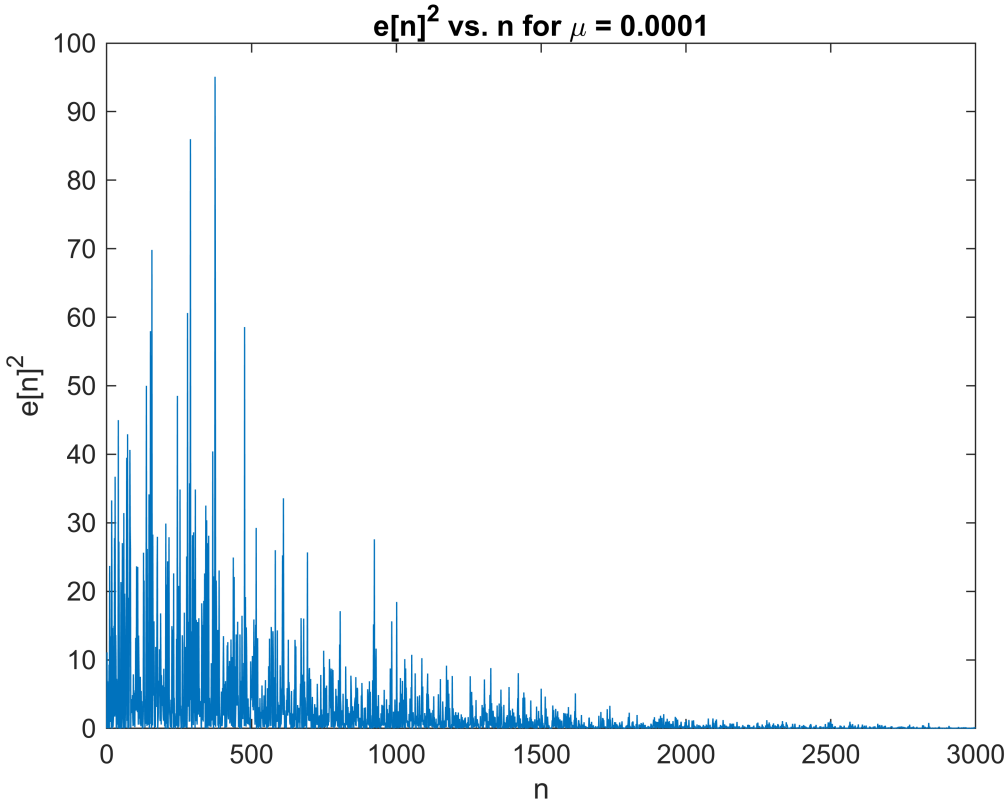
b) Repeat Problem 6a using an input of a*randn(1,N) with $a = \sqrt{10}$. What is the relationship between the adaptation constants which produce similar results here and in Problem 6a ?

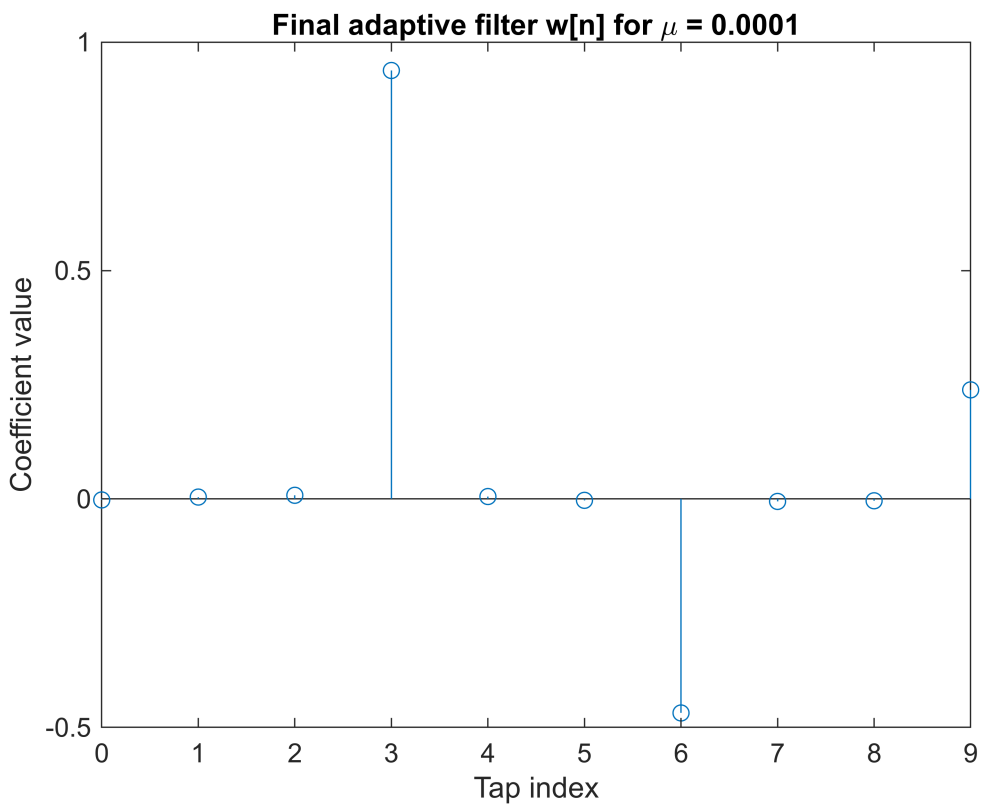
```
h = [1, 0, 0, -0.5, 0, 0, .25];
M = 10;
mu_vals = [1e-5, 1e-4, 1e-3];
N = 3000 + M - 1;
a = sqrt(10);
```

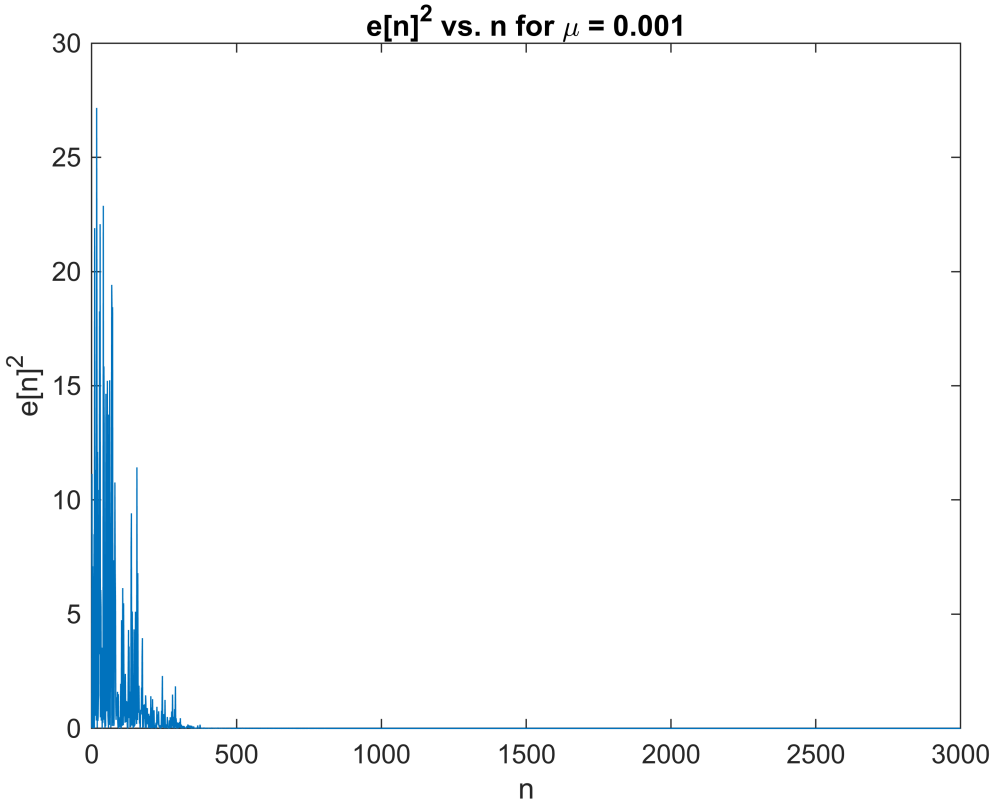
```
x = a*randn(1, N);

d = conv(x, h, 'valid');  % length = 3000

for k = 1:length(mu_vals)
    mu = mu_vals(k);

    w = zeros(M, 1); % adaptive filter
    e = zeros(1, 3000); % error
    y = zeros(1, 3000); % output


    for n = 1:3000
        x_n = x(n+M-1:-1:n).';
        y(n) = w' * x_n;
        e(n) = d(n) - y(n);
        w = w + mu * x_n * e(n);
    end

    % plot error squared
    figure;
    plot(0:2999, e.^2);
    title(['e[n]^2 vs. n for \mu = ', num2str(mu)]);
    xlabel('n');
    ylabel('e[n]^2');

    % stem plot of final filter coefficients
    figure;
    stem(0:M-1, w);
    title(['Final adaptive filter w[n] for \mu = ', num2str(mu)]);
    xlabel('Tap index');
    ylabel('Coefficient value');
end
```

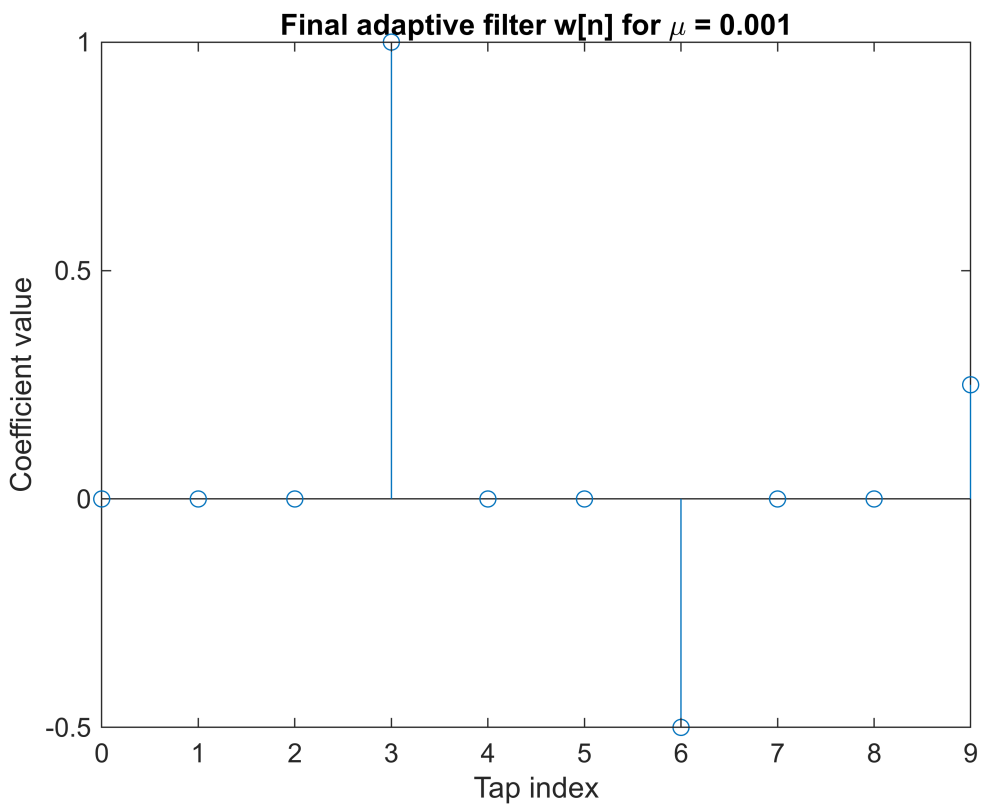

e[n]^2 vs. n for μ = 1e-05
e[n]^2
n
0
20
40
60
80
100
120
140
160
180
0
500
1000
1500
2000
2500
3000


Final adaptive filter w[n] for μ = 1e-05
Coefficient value
Tap index
0.25
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
0
1
2
3
4
5
6
7
8
9


e[n]^2 vs. n for μ = 0.0001
e[n]^2
n
0
10
20
30
40
50
60
70
80
90
100
0
500
1000
1500
2000
2500
3000


Final adaptive filter w[n] for μ = 0.0001
Coefficient value
Tap index
1
0.5
0
-0.5
0
1
2
3
4
5
6
7
8
9

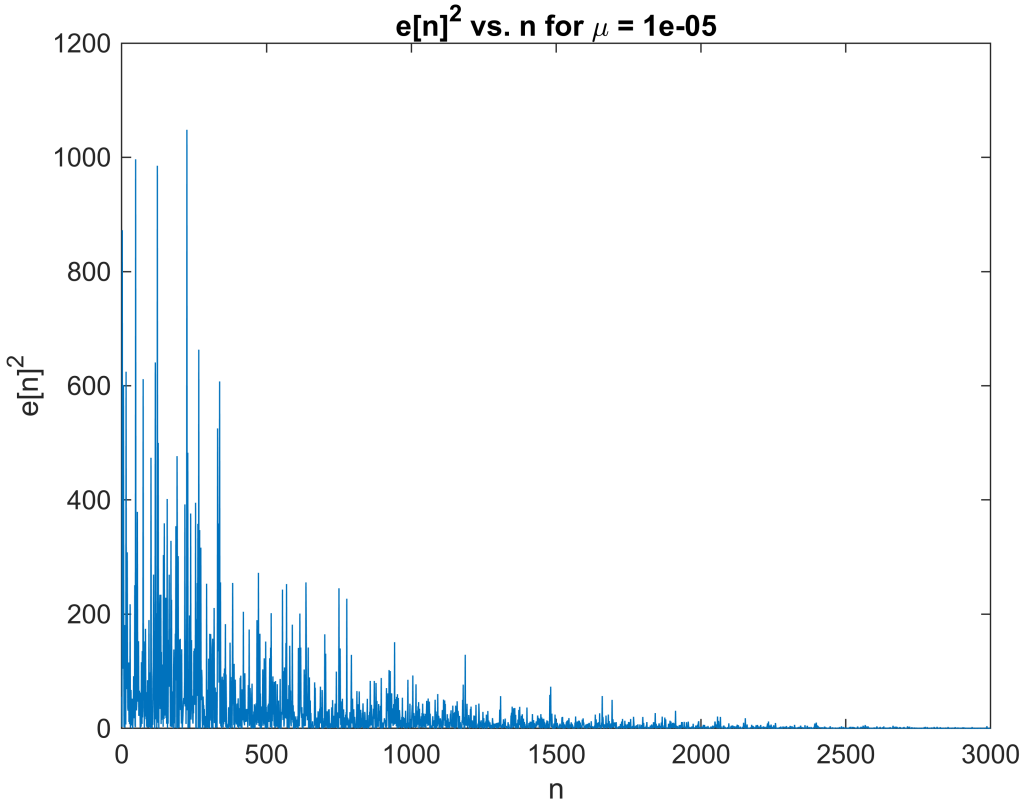


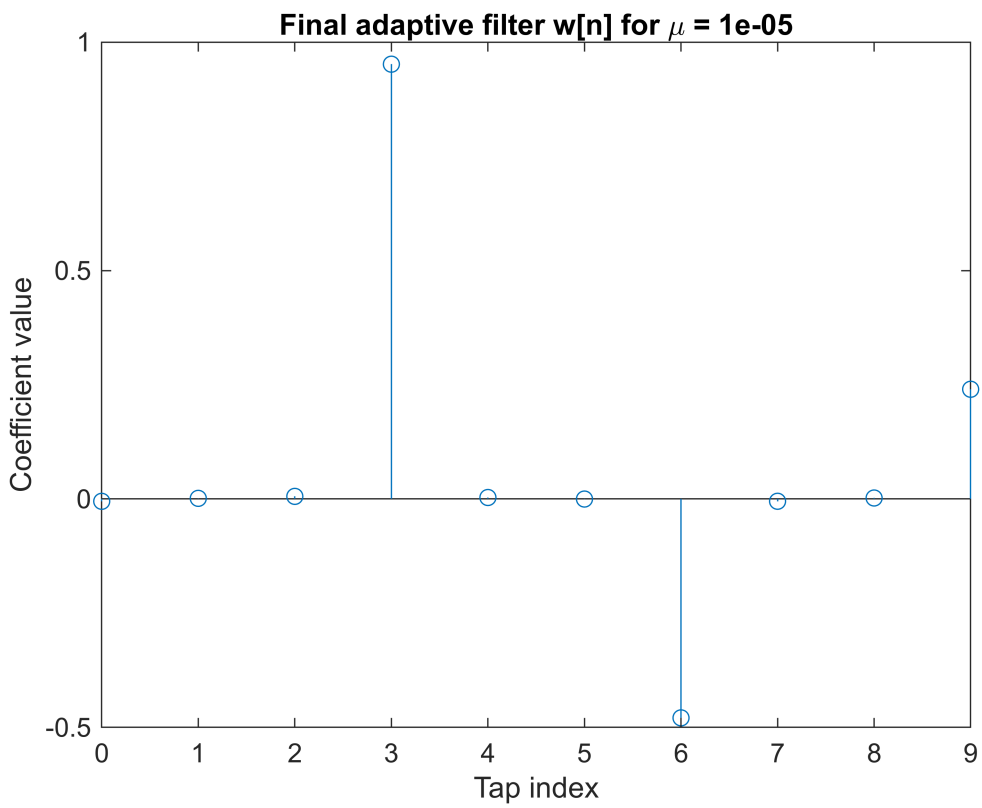


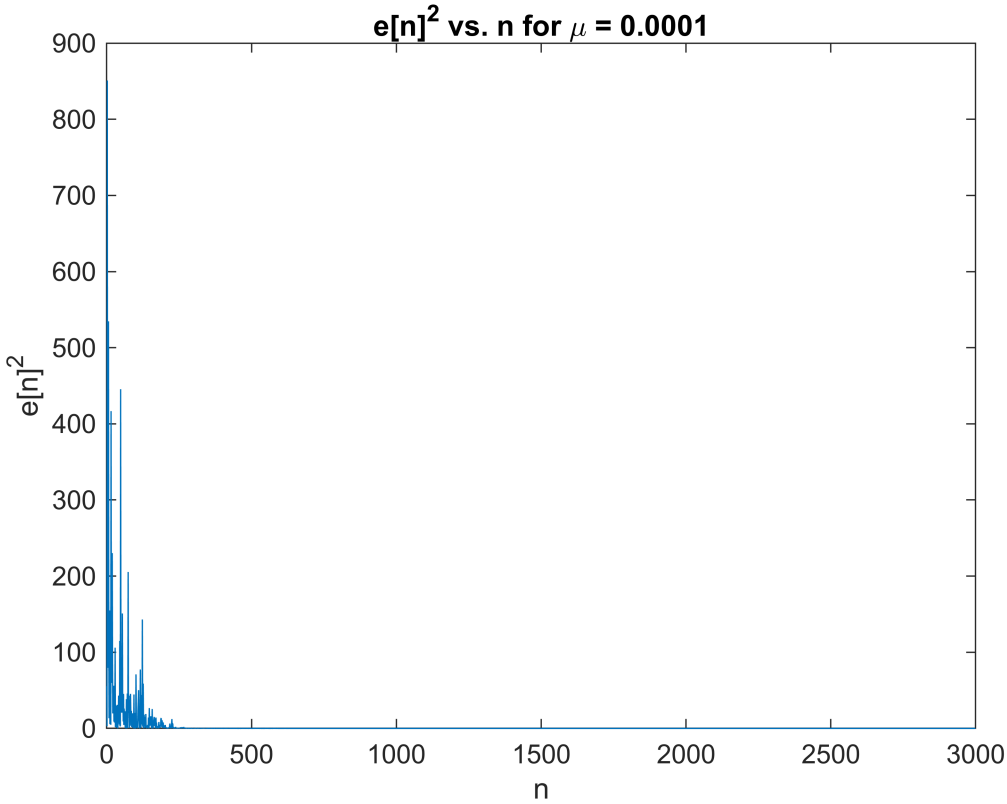
a helped the smaller mu values achieve more accurate weights and decrease the error function more rapidly. I presume the bigger the adaptation constant, the more accurate our weights can be.

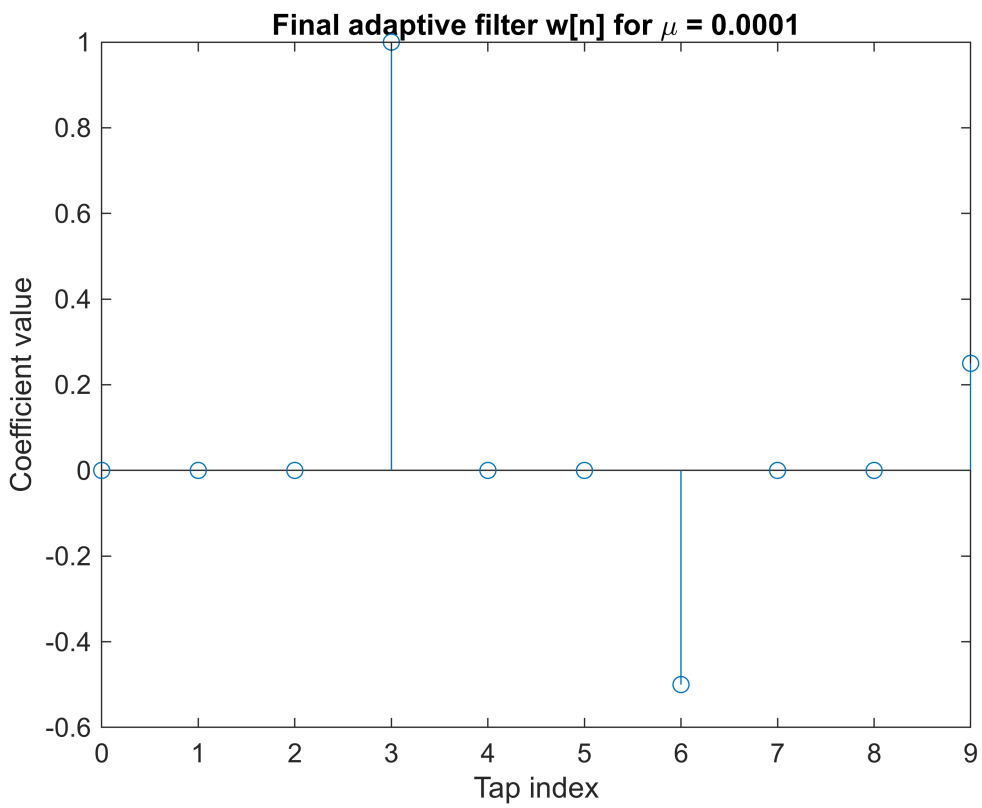
c) Repeat part (b) for $a = 10$.

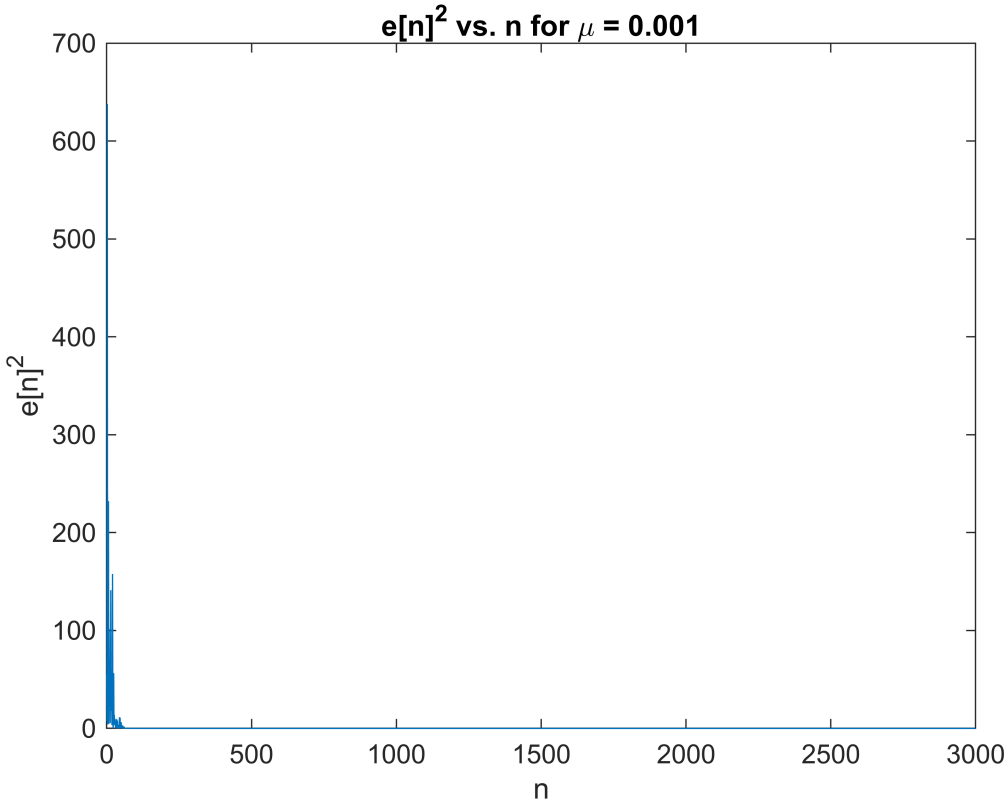
```
h = [1, 0, 0, -0.5, 0, 0, .25];
M = 10;
mu_vals = [1e-5, 1e-4, 1e-3];
N = 3000 + M - 1;
a = 10;

x = a*randn(1, N);

d = conv(x, h, 'valid');  % length = 3000

for k = 1:length(mu_vals)
    mu = mu_vals(k);

    w = zeros(M, 1); % adaptive filter
    e = zeros(1, 3000); % error
    y = zeros(1, 3000); % output


    for n = 1:3000
        x_n = x(n+M-1:-1:n).';
        y(n) = w' * x_n;
        e(n) = d(n) - y(n);
        w = w + mu * x_n * e(n);
    end

    % plot error squared
    figure;
    plot(0:2999, e.^2);
    title(['e[n]^2 vs. n for \mu = ', num2str(mu)]);
    xlabel('n');
    ylabel('e[n]^2');

    % stem plot of final filter coefficients
    figure;
    stem(0:M-1, w);
    title(['Final adaptive filter w[n] for \mu = ', num2str(mu)]);
    xlabel('Tap index');
    ylabel('Coefficient value');
end
```

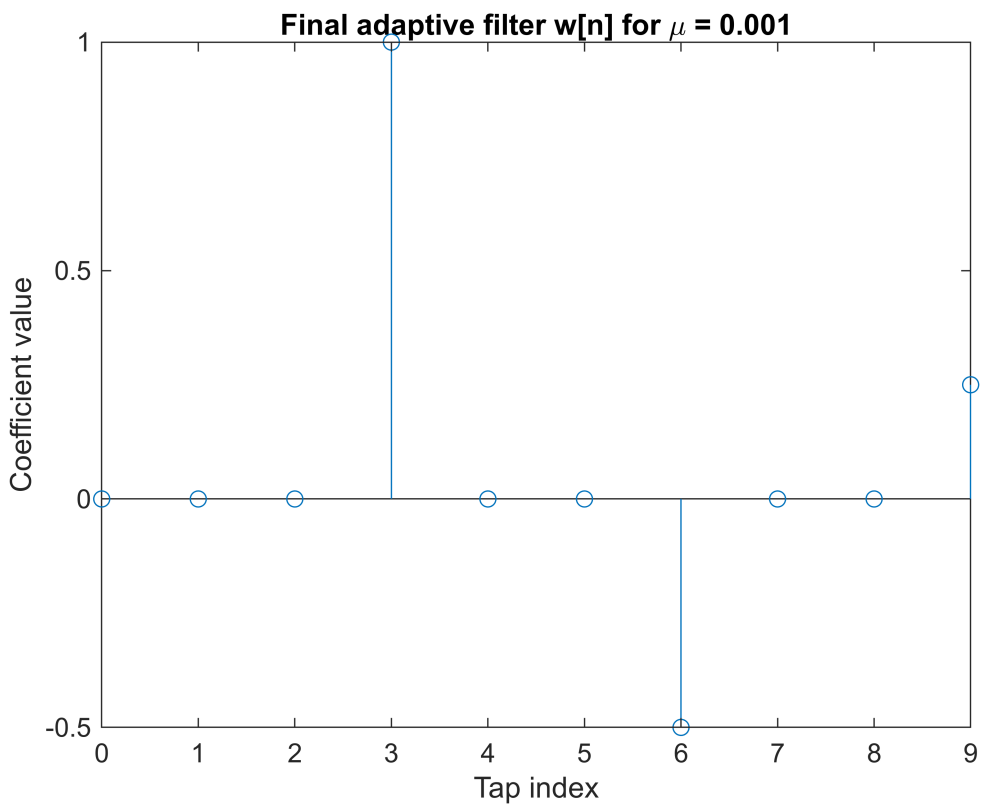

e[n]^2 vs. n for μ = 1e-05
1200
1000
800
600
400
200
0
e[n]^2
0
500
1000
1500
2000
2500
3000
n


Final adaptive filter w[n] for μ = 1e-05
1
0.5
0
-0.5
Coefficient value
0
1
2
3
4
5
6
7
8
9
Tap index


e[n]^2 vs. n for μ = 0.0001
e[n]^2
n
900
800
700
600
500
400
300
200
100
0
0
500
1000
1500
2000
2500
3000


Final adaptive filter w[n] for μ = 0.0001
Coefficient value
Tap index
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
0
1
2
3
4
5
6
7
8
9





As predicted, the higher adaptation constant led to more accurate weights and a much quicker dropoff on the error function output for all mus.